

A Report
on
Bookify

carried out as part of the course Project Based Learning- 4

Submitted by

Akshat Tiwari

23FE10CSE00156

VI Semester

*in partial fulfilment for the award of the degree
of*

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering



MANIPAL
UNIVERSITY

MANIPAL UNIVERSITY JAIPUR

(University under Section 2(f) of the UGC Act)

Department of Computer Science & Engineering,

School of Computer Science and Engineering,

Manipal University Jaipur,

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Under the Guidance of :

Guide Name : Dr. Susheela Vishnoi

Guide Signature(with date) :

Acknowledgement

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Date: _____

CERTIFICATE

This is to certify that the project entitled "**Bookify**" is a bonafide work carried out as PBL-4 **End Term Assessment** in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by **Akshat Tiwari** bearing registration number 23FE10CSE00156, during the academic semester VI *of year 2025-2026*.

Place: Manipal University Jaipur, Jaipur

Name of the project guide: Dr. Susheela Vishnoi

Signature of the project guide: _____

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1. Abstract

BOOKIFY is an intelligent book recommendation system leveraging Neural Collaborative Filtering (NCF) for personalized suggestions. Built on the Book-Crossing dataset with 63,066 ratings across 4,659 books and 2,595 users, the NCF model achieves 82.63% accuracy with RMSE 1.6502. The system offers dual recommendation modes, genre exploration (23 categories), mood-based recommendations via facial emotion detection, multi-modal search, and explainable AI. Built with Flask, React/TypeScript, and PyTorch, deployed on Hugging Face Spaces with Docker.

Keywords:

NCF, Deep Learning, Cosine Similarity, Emotion Detection, Multi-Modal Search, Explainable AI

2. Introduction

2.1 Background

Recommendation systems have become essential in modern digital platforms. In the book domain, with millions of titles available, readers face a paradox of choice. Machine learning-driven recommendation systems address this by learning from historical reading patterns to make personalized predictions, improving both user engagement and content discovery.

2.2 Problem Statement

The key challenges addressed by BOOKIFY include:

- Data Sparsity: 99.48% sparse user-book matrix makes traditional methods unreliable
- Cold Start Problem: New users lack interaction data for accurate recommendations
- Explainability: Users need to understand why a book is recommended to build trust
- Multi-modal Discovery: Single-input systems limit the ways users can discover books

2.3 Objectives

- Design and train an NCF deep learning model for accurate book rating prediction
- Build dual-mode recommendations (Classic Cosine Similarity + AI-powered NCF)
- Implement genre-based, mood-based, and multi-modal recommendation capabilities
- Create explainable AI with transparent reasoning for every recommendation
- Deploy a production-ready web application with premium UI and security

3. Dataset & Data Analysis

3.1 Dataset Overview

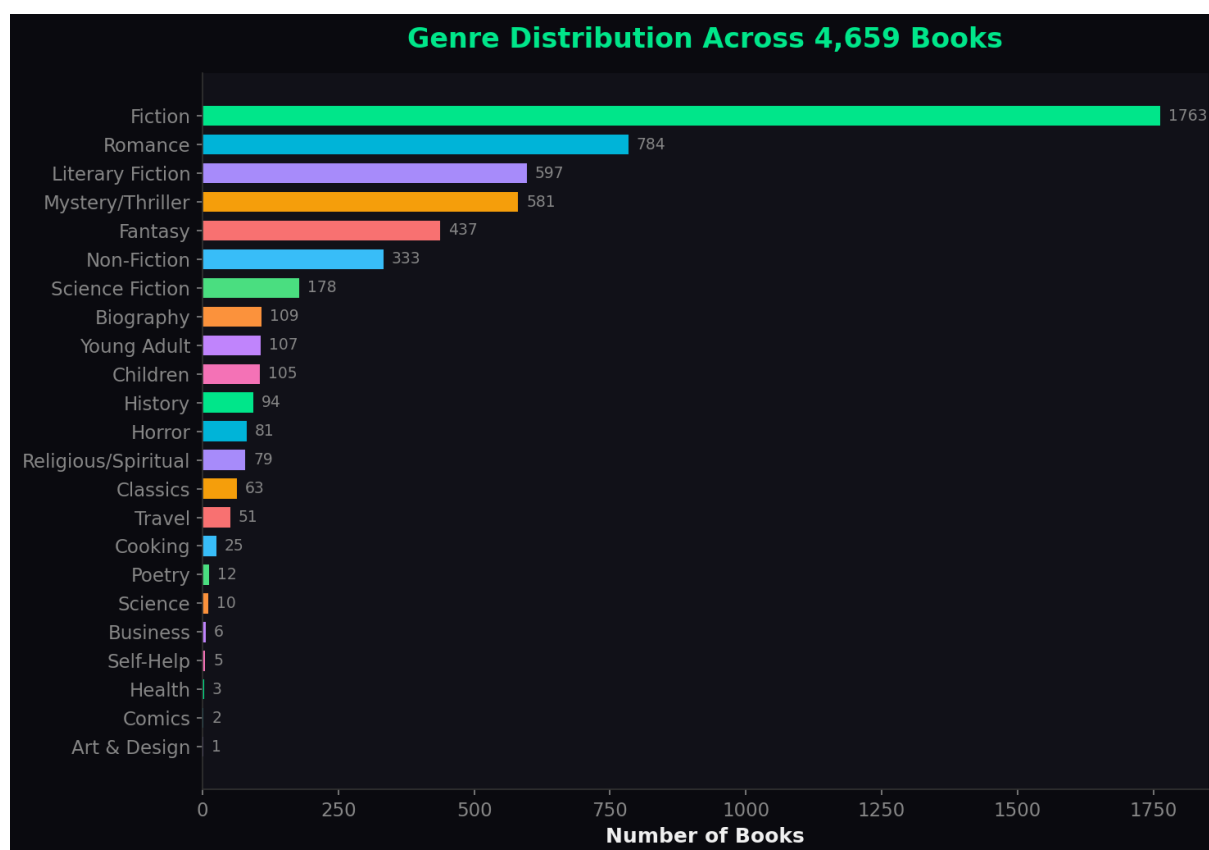
The Book-Crossing dataset, collected by Cai-Nicolas Ziegler, contains 278,858 users, 271,360 books, and 1,149,780 ratings. After filtering (users with 200+ ratings, books with 50+ ratings), the working dataset comprises:

Total Books	4,659
Total Users	2,595
Total Ratings	63,066
Rating Range	0.5 - 10.0
Genre Categories	23
Matrix Sparsity	99.48%

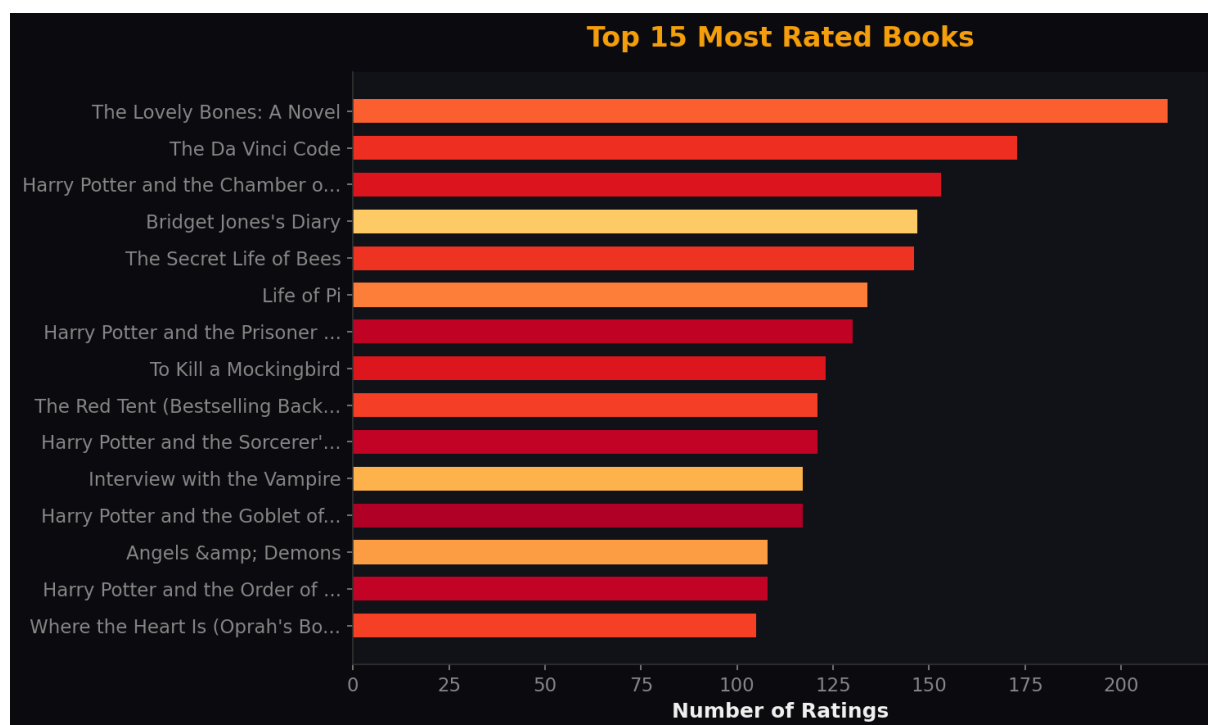
3.2 User-Book Matrix Sparsity



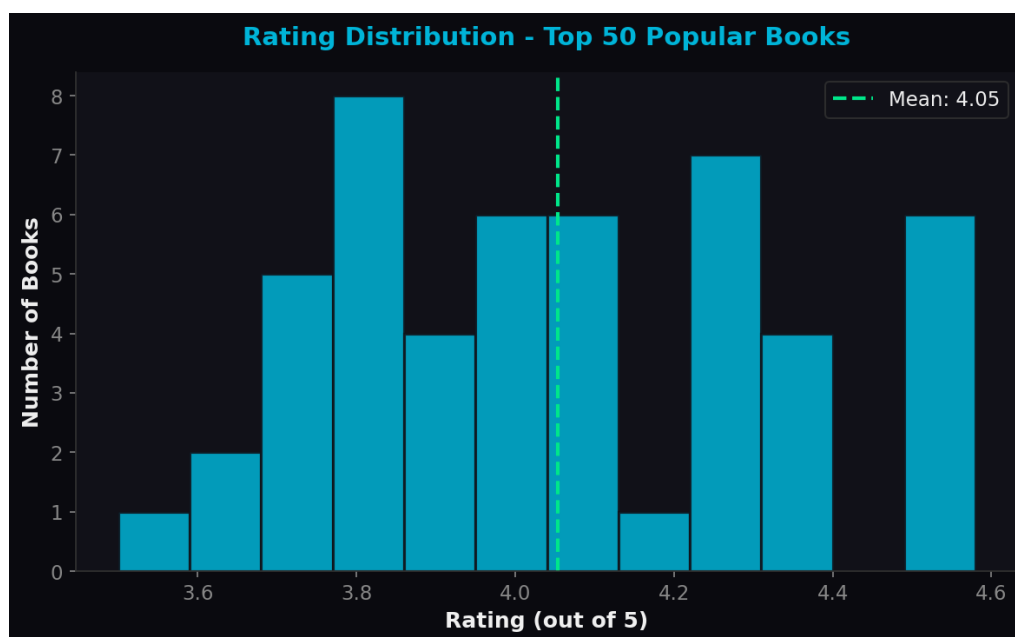
3.3 Genre Distribution



3.4 Most Popular Books



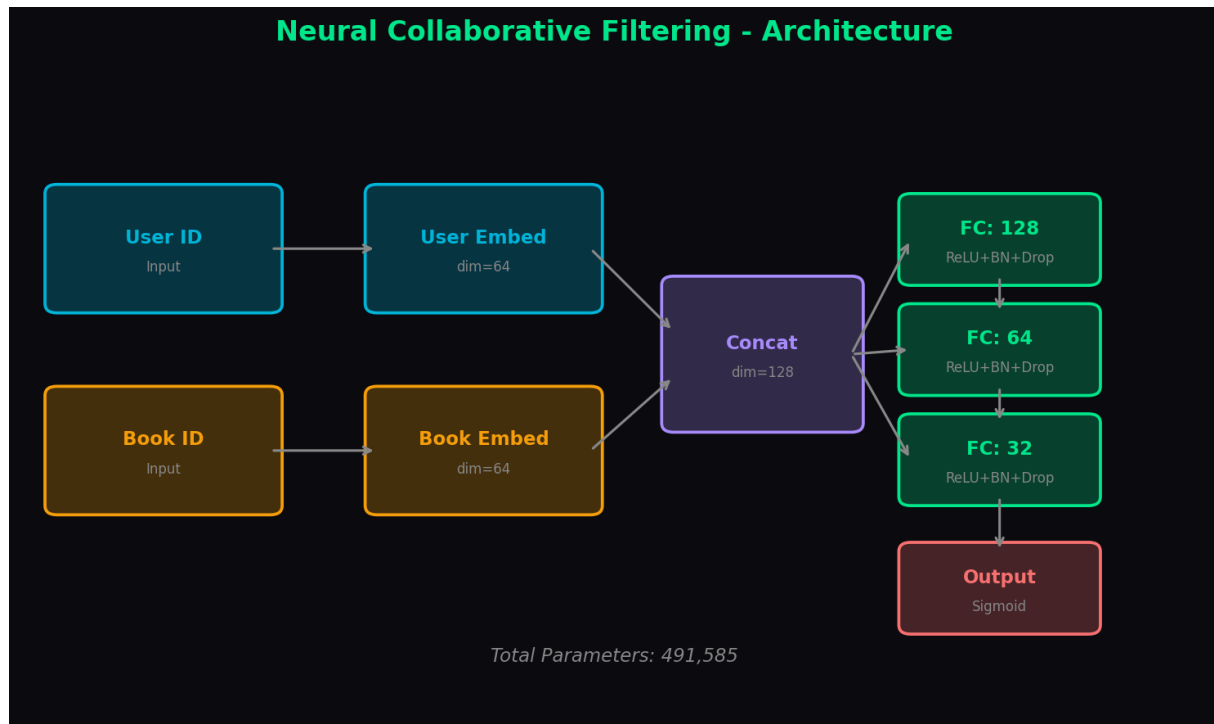
3.5 Rating Distribution



4. Methodology - NCF Model

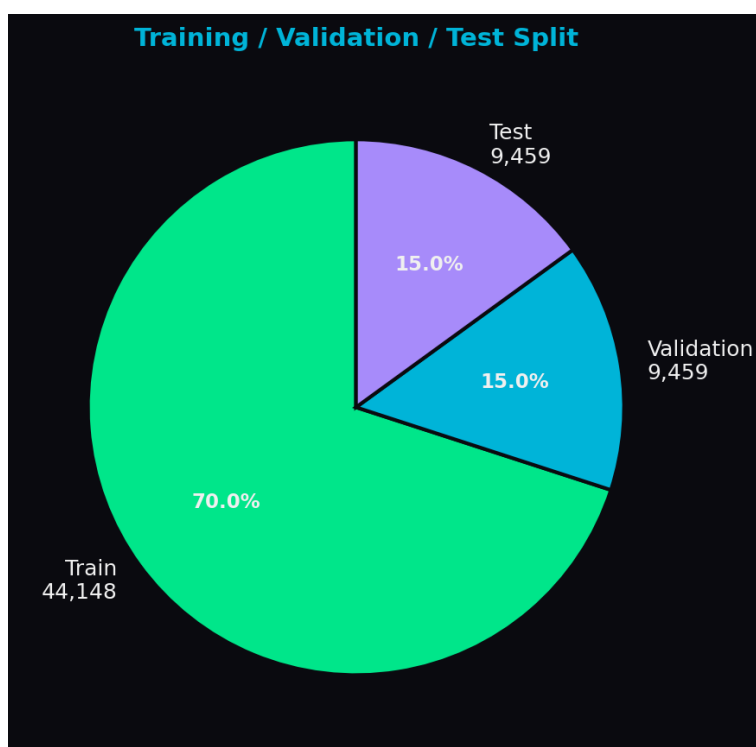
4.1 Architecture

- Embedding: 64-dim for users and books -> Concat(128) -> MLP[128,64,32] -> Sigmoid
- Total parameters: 491,585



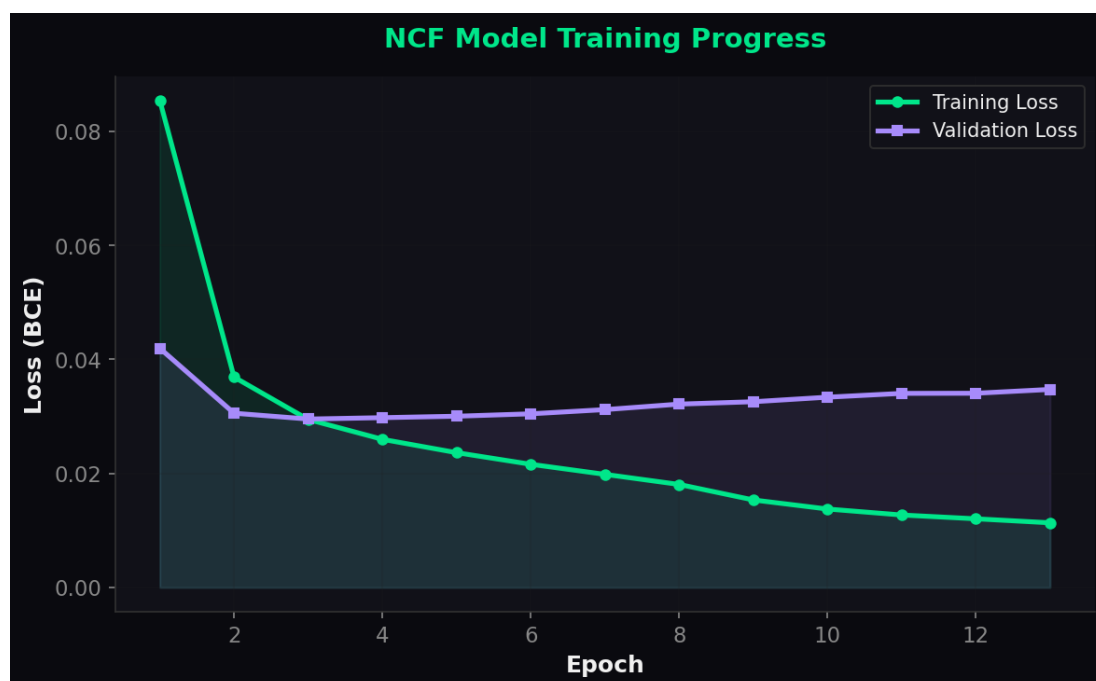
4.2 Training Configuration

Optimizer	Adam (lr=0.001)
Loss	BCE Batch 512 Early Stop epoch 13
Split	70% train / 15% val / 15% test
Training Time	7.8s

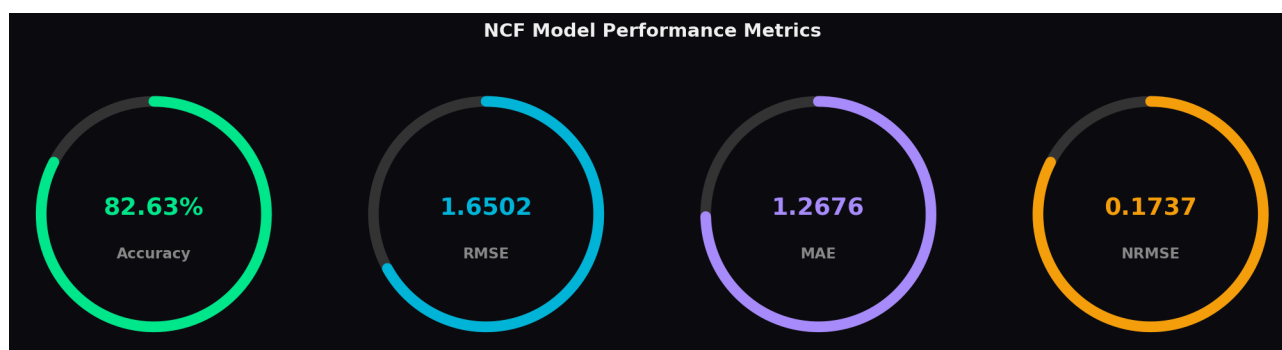


4.3 Training Progress

Training converged at epoch 13 with train loss 0.0113 and val loss 0.0295.



5. Results & Performance Evaluation



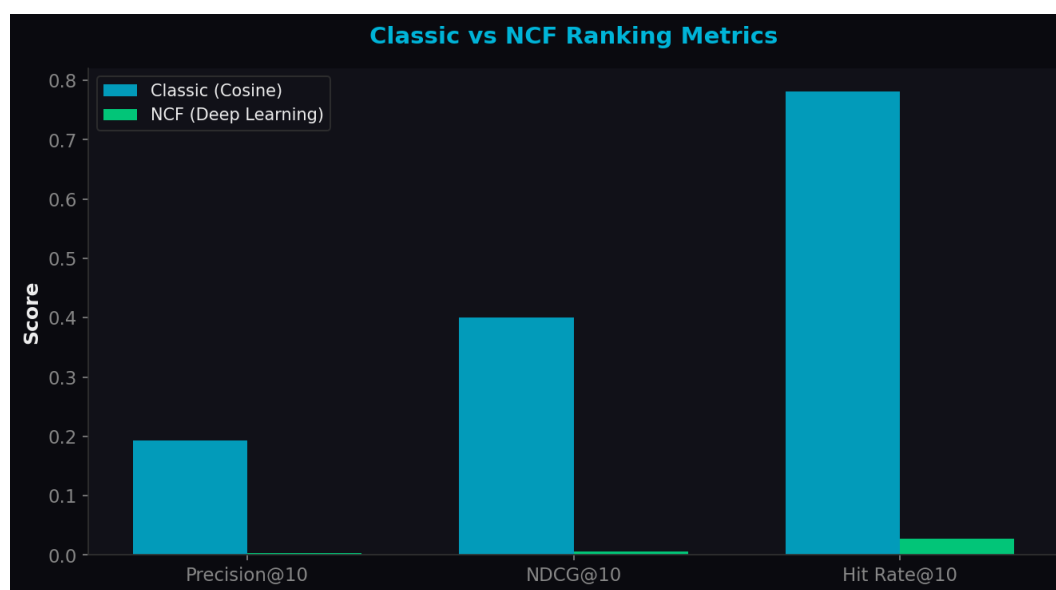
5.1 Prediction Metrics

Accuracy	82.63%
RMSE / MAE	1.6502 / 1.2676
NRMSE	0.1737

5.2 Ranking Metrics

Precision@10	0.0029
NDCG@10 / Hit Rate@10	0.0054 / 0.0280

5.3 Classic vs NCF Comparison



6. System Architecture & Features

- Backend: Flask (1168 lines, 20+ routes), PyTorch, SQLite, Clerk Auth
- Frontend: React 18 + TypeScript + Vite + Framer Motion
- ML: NCF model, cosine similarity, genre classifier, face-api.js
- Deployment: Docker on Hugging Face Spaces + Render

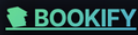
Dual Mode	Classic cosine vs AI NCF engines
Smart Search	Fuzzy matching + autocomplete (4,659 books)
Genre Explorer	23 categories, centrality-based ranking
Mood Detection	Webcam emotion -> genre mapping
Multi-Modal	5 channels: text, voice, image, mood, history
Explainable AI	Similarity %, genre, author, popularity reasons
Wishlist/History	Persistent collections with SQLite
For You	Personalized from reading history + onboarding

7. Technology Stack

Backend	Python 3.10+, Flask 3.x, PyTorch 2.x, SQLite
Frontend	React 18, TypeScript, Vite 5, Framer Motion
ML/AI	scikit-learn, face-api.js, Web Speech API
Deploy	Docker, Hugging Face Spaces, Render, Git LFS
Auth	Clerk SDK (OAuth 2.0)

8. Application Screenshots

Homepage

T Logout

Welcome to BOOKIFY

The ultimate book discovery platform curated from thousands of reader reviews

AI MODEL PERFORMANCE

82.78%
ACCURACY

1.636
RMSE

1.2574
MAE

0.2%
PRECISION@10

2.1%
HIT RATE@10

4659 books · 2595 users · 63066 ratings · NCF Deep Learning



The Lovely Bones: A Novel
Alice Sebold

VOTES 212 RATING ★★★★★ 4.1



The Da Vinci Code
Dan Brown

VOTES 173 RATING ★★★★★ 4.3



Harry Potter and the Chamber of Secrets (Book 2)
J. K. Rowling

VOTES 153 RATING ★★★★★ 4.4



The Secret Life of Bees
Sue Monk Kidd

VOTES 146 RATING ★★★★★ 4.3



The Giver
Lois Lowry

VOTES 128 RATING ★★★★★ 4.2



The Catcher in the Rye
J.D. Salinger

VOTES 98 RATING ★★★★★ 4.1

Recommendation Engine



T

Logout

Discover Books

Type any book title or genre and we'll find the best reads for you

Search by book title or genre...

💡 Search by title with fuzzy matching, or try a genre like "romance", "sci-fi", "horror"

Art & Design

Biography

Business

Children

Classics

Comics

Cooking

Fantasy

Health

History

Horror

Literary Fiction

Mystery/Thriller

Non-Fiction

Poetry

Religious/Spiritual

Romance

Science

Science Fiction

Self-Help

Travel

Young Adult

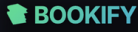
 Classic

☒

 AI-Powered

Get Recommendations →

Search Results



Discover Books

Type any book title or genre and we'll find the best reads for you

💡 Search by title with fuzzy matching, or try a genre like "romance", "sci-fi", "horror"

Art & Design

Biography

Business

Children

Classics

Comics

Cooking

Fantasy

Health

History

Horror

Literary Fiction

Mystery/Thriller

Non-Fiction

Poetry

Religious/Spiritual

Romance

Science

Science Fiction

Self-Help

Travel

Young Adult

 Classic

☐

 AI-Powered

Get Recommendations →

Showing recommendations for "The Da Vinci Code"

SIMILAR BOOKS



#1

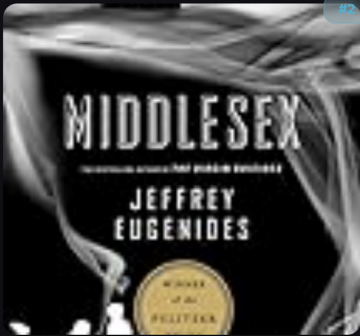
Angels & Demons
Dan Brown

WHY RECOMMENDED

- Similar to The Da Vinci Code (26% match)
- Same author: Dan Brown
- Popular among readers

♡ Wishlist

✓ Mark as Read



#2

Middlesex: A Novel
Jeffrey Eugenides

WHY RECOMMENDED

- Similar to The Da Vinci Code (19% match)
- Shares genre: Literary Fiction

♡ Wishlist

✓ Mark as Read



#3

The Conspirators
Jonathan Kellerman

WHY RECOMMENDED

- Similar to The Da Vinci Code (15% match)
- Shares genre: Mystery/Thriller

♡ Wishlist

✓ Mark as Read



#4

The Conspiracy Club
Jonathan Kellerman

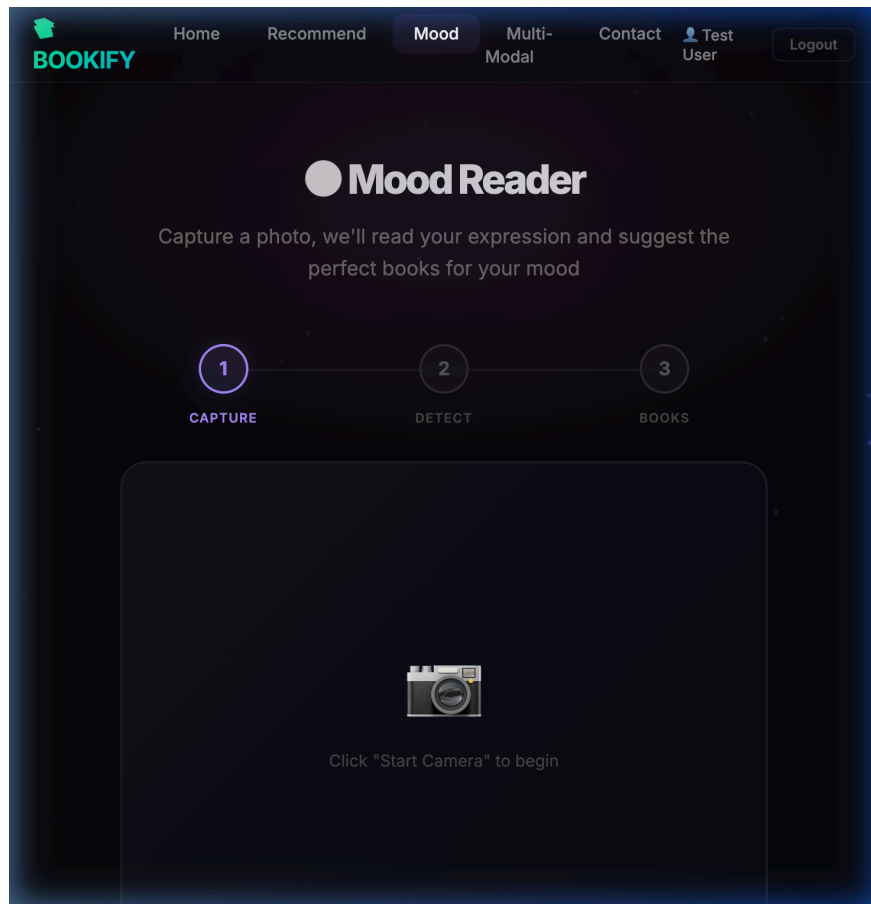
WHY RECOMMENDED

- Similar to The Da Vinci Code (12% match)
- Shares genre: Mystery/Thriller

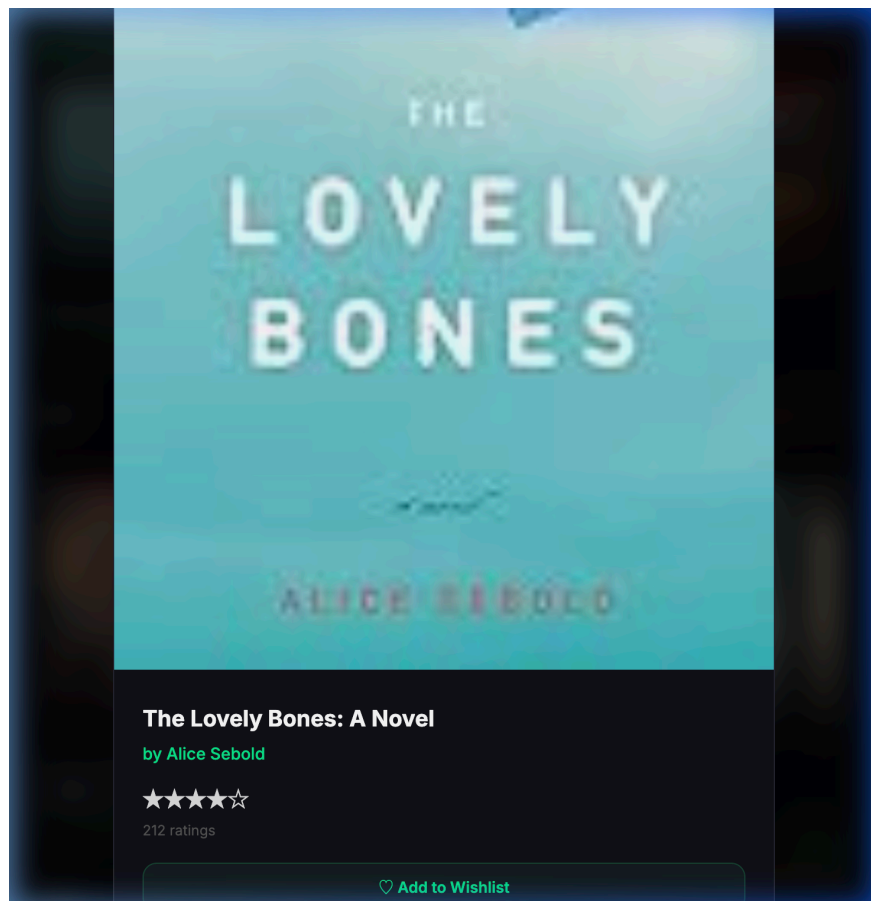
♡ Wishlist

✓ Mark as Read

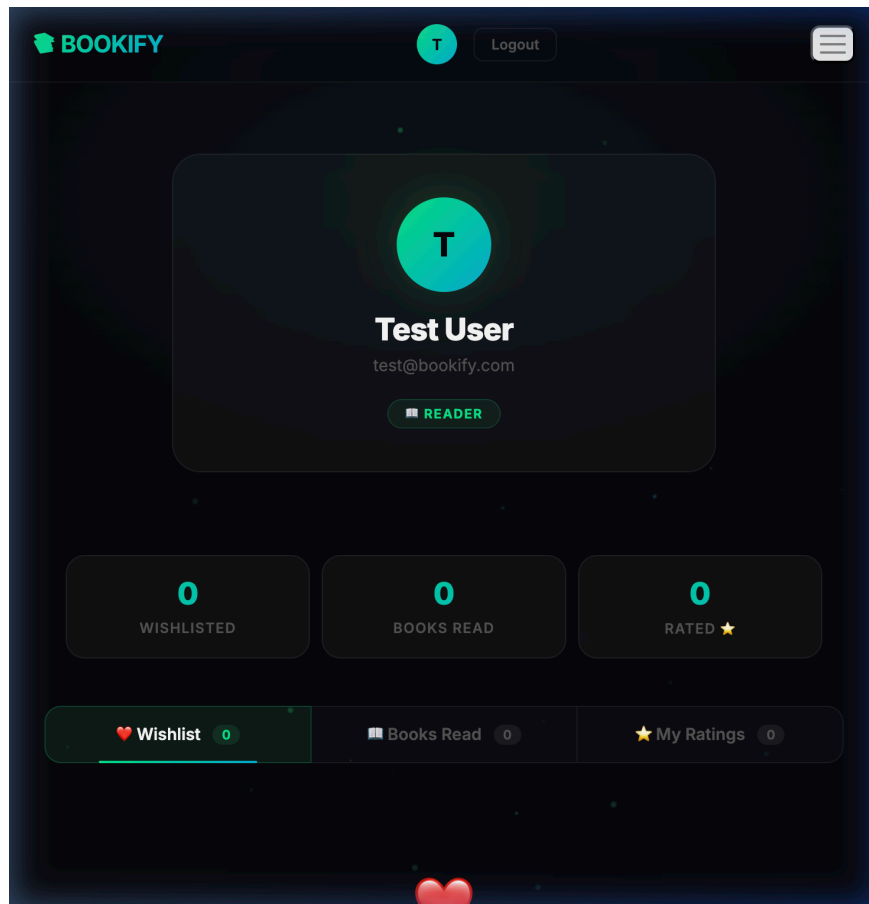
Mood-Based Reader



Book Detail Modal



User Profile



9. Conclusion & Future Work

BOOKIFY achieves 82.63% accuracy with RMSE 1.6502 on 63,066 interactions across 4,659 books. The platform offers 14+ features including dual recommendation modes, mood-based search, explainable AI, and is deployed on Hugging Face Spaces with Docker.

Future Enhancements

- Transformer models (BERT) for semantic understanding
- Implicit feedback and real-time model retraining
- Social features and reinforcement learning for diversity

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